

Welcome to the GenAEx 6.5 website!



GenAEx 6.5 Overview

GenAEx 6.5 (Peakall and Smouse 2006, 2012) offers a wide range of population genetic analysis options for the full spectrum of genetic markers within the Microsoft Excel environment on both PC and Macintosh computers. When combined with its user-friendly interface, rich graphical outputs for data exploration and publication, tools for data manipulation and export options to many other software packages, we believe that GenAEx offers an ideal launching pad for population genetic analysis by students, teachers and researchers alike.

Publications about GenAEx

Peakall, R. and Smouse P.E. (2012) GenAEx 6.5: genetic analysis in Excel. Population genetic software for teaching and research-an update. *Bioinformatics* 28, 2537-2539.

Freely available here: <http://bioinformatics.oxfordjournals.org/content/28/19/2537>

***Peakall, R. and Smouse P.E. (2006) GENALEX 6: genetic analysis in Excel. Population genetic software for teaching and research. *Molecular Ecology Notes* 6, 288-295.**

*In June 2018, Google Scholar indicates the number of citations for GenAEx exceeds 14,000! We thank the many users for using and citing GenAEx!

GenAEx 6.51b2 - Released June 2018

In this beta release of GenAEx 6.51 we offer the new QDiver analysis option introduced in our publication:

Smouse, P. E., Banks, S. C., and Peakall, R. (2017) Converting quadratic entropy to diversity: Both animals and alleles are diverse, but some are more diverse than others. *PLOS ONE* 12, e0185499.

Freely available here: <http://journals.plos.org/plosone/article?id=10.1371/journal.pone.0185499>

GenAEx 6.503 - Released Dec 2016

This release for Excel 2016 running in Windows offered new access to all GenAEx options via the Excel Ribbon, while at the same time remaining backwards compatible to Excel 2010. Prior to this release, all access to GenAEx options was via the GenAEx menu. The location of the GenAEx menu has depended on the version of Excel used, and the operating system. The Excel Ribbon now provides a common interface and access point to GenAEx options across Excel versions and operating systems.

Recent tests indicate Visual Basic for Applications (VBA) functionality on the Macintosh is increasingly being improved by Microsoft. Consequently, most GenAEx options can now be run in GenAEx 2016 on the Macintosh via the Ribbon, although some minor Dialog box and Ribbon button image issues remain. These minor, mostly cosmetic issues, have yet to be fully resolved. We apologize to our long-standing GenAEx users on the Macintosh for these issues which are beyond our control.

Note that in 2013, GenAEx 6.501 was released as a critical update for all GenAEx users! Please read the Release History and Read Me pdfs contained in the download pack for crucial information about the improvements and bug fixes.

More About GenAEx 6.5

GenAEx: Genetic Analysis in Excel is a popular cross platform package for population genetic analysis that runs within Microsoft Excel. GenAEx offers analysis of codominant, haploid and binary genetic loci and DNA sequences. Both frequency-based (F-statistics, heterozygosity, HWE, population assignment, relatedness) and distance-based (AMOVA, PCoA, Mantel tests, multivariate spatial autocorrelation) analyses are provided. In GenAEx 6.5 we introduce exciting new features including calculation of new estimators of population structure: G'_{ST} , G''_{ST} , Jost's D_{est} , and F'_{ST} via AMOVA, Shannon Information analysis, linkage disequilibrium analysis for biallelic data, and heterogeneity tests for spatial autocorrelation analysis. Direct data export is provided to more than 30 other software packages, and indirectly via common formats to many more packages. New teaching tutorials and extra step-by-step options are also included. The comprehensive independently written guide has been fully revised.

Go to [Download](#) to obtain GenAEx 6.503 or to test the beta release of GenAEx 6.51!